

8

Force and Laws of Motion



While pulling a cart, the force is applied in the direction of a person whereas pull is applied in the direction away from the person. There are more examples of push and pull like opening a drawer, drawing a bucket of water from the well.



What's Inside

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- ✓ Case-based Examples

Questions

- ✓ New Pattern Qs
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- ✓ Detailed Explanation

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| TOPIC 1 |

DESCRIPTION OF FORCE

Learning Objectives

- Students will learn how to define the force.
- Students will learn about the details of the effects of force.

Learning Outcomes

- Students will be able to explain the term force.
- Students will be able to appreciate the effects of force from day-to-day examples.

Real Life Application

Do you love watching or playing cricket? If yes, then you must probably know the rules of this game. Cricket is one of the most prevalent sports in the world.

You must have seen that players apply different batting techniques to hit the ball. The impact between bat and ball is an extremely violent one where the bat imparts a huge force on the ball, causing it to change direction and gain speed. If a batsman does not apply enough force on the ball, it cannot travel the long distance in the air making the player lose his runs.



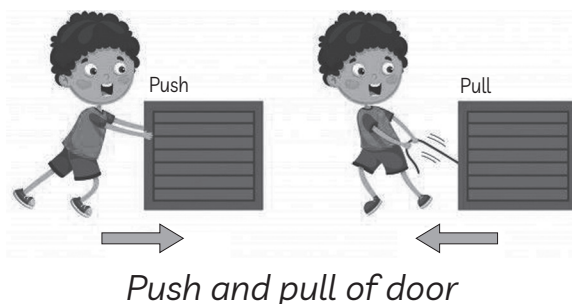
The average force acting on the ball can be estimated by using Newton's law of motion which you will study in detail in this chapter.

This is how the force plays an important role in the field of sports. Force is applied in many other places in our day-to-day activities such as while pushing, pulling or lifting any object. Now let us learn more about force.

In our everyday actions like lifting boxes, pushing and pulling doors, stretching rubber bands, twisting the lid to open jars and pressing rubber balls, all these things happen when we apply some force. Similarly, we see that moving a stationary object or stopping a moving object requires some force.



Force is an external effect which changes or tends to change the state of rest, motion, direction or shape of an object. Force cannot be seen but its effect can be felt according to the force acting on an object.



Lifting the box



Twisting the lid to open jars

Effects of Force

The various effects of force are as follows:

- (1) **Force can make the object move or stop:** When a cricket ball which is at rest is hit by a bat, then the ball comes into motion. Force is applied to the bat that brings the ball in motion. It starts rolling, but as the hand is placed in front of the moving ball, the moving ball comes to rest. A force applied to a moving ball by a handheld in its path, brings it to rest.



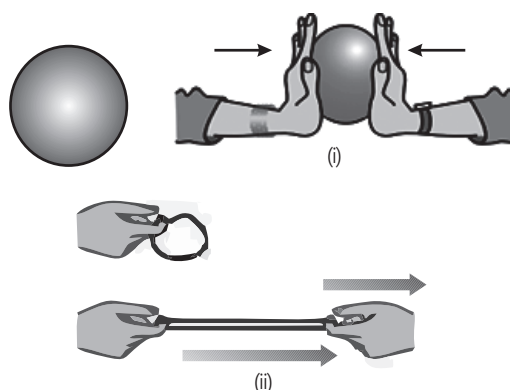
- (2) **Force can change the direction of a moving object:** When the football is kicked by our foot, then the direction of the ball changes. In this case, the force exerted by the foot changes the direction of football.



- (3) **Force can change the speed of the moving body:** While riding the bicycle at a certain speed, someone pushes the moving bicycle from behind and then the speed of the bicycle increases. Similarly, if someone pulls the moving bicycle from behind, then the speed of the bicycle decreases. Hence, the push or pull changes the speed of the body.



- (4) **Force can change the size and shape of a body:** As the force is applied to the spherical rubber ball, it becomes oblong. Hence, the force changes the shape of the body.



When the rubber band is stretched, the rubber band elongates and its shape changes.

Types of Forces

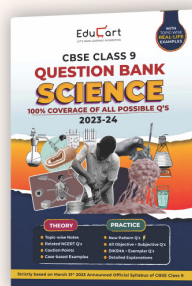
Force is categorised in to two forms:

- (1) Balanced forces
- (2) Unbalanced forces



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OBJECTIVE Type Questions

[1 mark]

Multiple Choice Questions

1. A ball was rolled on a frictionless horizontal surface with velocity 5 m/s having a mass of 100 gm. Find the force required to keep moving the ball with the same velocity.

(a) 500 N (b) 0.5 N
(c) 0.05 N (d) 0 N

Ans. (d) 0 N

Explanation: Given, mass = 100 gm = 0.1 kg

Velocity, $v = 5$ m/s

As the ball is moving with uniform velocity on a frictionless surface.

So, acceleration, $a = 0$ m/s²

As we know, Force = mass $\times$ acceleration

$$F = 0.1 \times 0 = 0$$

As the surface is frictionless, no external force acts on the ball, it will remain in its original state of motion.



Related Theory

→ According to the property of inertia, if no external force acts on the body, it will stay in its current state, i.e., if it is at rest, it will stay at rest and if it is in motion, it will stay in motion. Since friction opposes force, no force is necessary to keep the item moving uniformly.

2. The sum of momentum of the two objects before collision is equal to the sum of momentum after the collision provided there is no external unbalanced force acting on them. This is known as the law of conservation of momentum. This statement can alternatively be given as the total momentum of the two objects is unchanged or conserved by the collision. Law of conservation of momentum is applicable to system of particle. Answer the following questions.

Law of conservation of momentum is applicable to

(a) System of particles
(b) Only for 2 particles
(c) Only for 1 particle
(d) None of the above

[Delhi Gov. SQP 2022]



Ans.(a) *System of particles*

Explanation: Law of conservation of momentum states that “two bodies acting upon each other in an isolated system will have a constant total momentum until an external force is applied upon them. The motion never changes in an isolated collection of objects and the total momentum of a system remains constant. Hence, it is applicable to a system of particles and not to individual bodies.

Assertion-Reason (A-R)

This question has two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

- (a) Both A and R are true and R is the correct explanation of A.
 - (b) Both A and R are true and R is not the correct explanation of A.
 - (c) A is true but R is false.
 - (d) A is false but R is true.
3. Assertion (A): If a large number of concurrent forces act at the same position on an object, the object is in equilibrium if the sum of all forces equals zero.

Reason (R): In mechanics, equilibrium of a particle refers to a situation in which the particle's net external force is zero.

Ans. (a) *Both A and R are true and R is the correct explanation of A.*

Explanation: Concurrent forces are those in which all of the forces' axis of action cross at the same position. The object will be in equilibrium if the sum of all forces is zero, as the resultant or net force is zero.

The term "equilibrium" refers to a situation in which the sum of the forces equals zero.

Fill in the Blanks

4. Complete this sentence to describe Newton's First Law.

An object in constant remains in unless a resultant acts on it. [British Council 2022]

Ans. *rest or motion, rest or motion, unbalanced force*

Explanation: According to Newton's first law of motion, "An object at rest stays at rest, and an object in motion stays in motion with the same speed and direction unless acted upon by an unbalanced force".

True or False

5. An object is said to be in a state of equilibrium if it is moving with constant speed.

Ans. True

Explanation: If the net force acting on a body is zero, it is considered to be in equilibrium. According to Newton's second law, the acceleration of such objects will be zero. A constant velocity implies zero acceleration.



Related Theory

→ Acceleration is defined as the rate of change of velocity. It is expressed as:

$$a = \frac{v-u}{t}$$

Match the Following

6. Match the column I with column II.

Column I	Column II
(A) Force	(i) Resist the change in state of an object
(B) Acceleration	(ii) kg-m/s
(C) Momentum	(iii) m/s ²
(D) Inertia	(iv) kg-m/s ²

Codes:

(a) A-(iii), (B)-(ii), C-(iv), D-(i)

(b) A-(iv), B-(iii), C-(ii), D-(i)

(c) A-(iv), B-(ii), C-(iii), D-(i)

(d) A-(iii), B-(i), C-(iv), D-(ii)

Ans. (b) A-(iv), B-(iii), C-(ii), D-(i)

Explanation: Force

$$\begin{aligned} &= \text{mass} \times \text{acceleration} \\ &= \text{kg-m/s}^2 = \text{Newton (N)} \end{aligned}$$

Acceleration

$$\begin{aligned} &= \frac{\text{Change in velocity}}{\text{Time}} \\ &= \frac{\text{m/s}}{\text{s}} = \text{m/s}^2 \end{aligned}$$

Momentum = mass x velocity = kg-m/s

Inertia resists the change in state of rest or motion in a straight line until an external force is applied.

CASE BASED Questions (CBQs)

[4 & 5 marks]

Read the following passages and answer the questions that follow:

7. While going on a trip, Rahul found that the taxi was moving along a straight line with a uniform velocity of 108 km/h. The velocity of the taxi slows down to 72 km/h in 4 seconds by an external force. (The mass of the taxi is 600 kg.)



- (A) Find the initial velocity and final velocity in metres per second.
(B) Find the change of momentum.
(C) How much external force is applied to the taxi?

Ans.(A) Conversion of km/h to m/s is:

$$1 \text{ km/h } \frac{5}{18} = \text{m/s}$$

$$\text{Initial velocity, } u = 108 \text{ km/h}$$

$$u = 108 \times \frac{5}{18} = 30 \text{ m/s}$$

$$\text{Final velocity, } v = 72 \text{ km/h}$$

$$v = 72 \times \frac{5}{18} = 20 \text{ m/s}$$

(B) Initial momentum, $p_1 = mu$

$$p_1 = 600 \times 30 = 18000 \text{ kg-m/s}$$

Final momentum, $p_2 = mv$

$$p_2 = 600 \times 20 = 12000 \text{ kg-m/s}$$

$$\begin{aligned} \text{Net Change in momentum, } p &= p_2 - p_1 \\ &= 12000 - 18000 \\ &= -6000 \text{ kg-m/s} \end{aligned}$$

(C) Acceleration is defined as the rate of change of velocity.

$$a = \frac{v-u}{t}$$

$$a = \frac{20-30}{5} = -2 \text{ m/s}^2$$



Negative signs show retardation.

$$F = ma$$

$$F = 600 \times 2$$

$$F = 1200 \text{ N}$$

The external force applied is 1200 N

VERY SHORT ANSWER Type Questions (VSA)

[1 mark]

8. A body of mass 5 kg is moving with an acceleration of 5 m/s^2 . What is the rate of change of momentum? [Diksha]

Ans. Given, mass, $m = 5 \text{ kg}$

Acceleration, $a = 5 \text{ m/s}^2$

According to Newton's second law of motion

Rate of change of momentum = force applied,

$$F = ma$$

$$= 5 \times 5 \text{ kg-m/s}^2 = 25 \text{ N}$$

9. Heavier bodies require greater effort to put them in motion as compared to lighter bodies. Why? [Diksha]

Ans. According to Newton's second law of motion,

$$F = ma.$$

More the mass, more will be the inertia hence greater force will be required to move an object.

SHORT ANSWER Type-I Questions (SA-I)

[2 marks]

10. Which of the following has more inertia?

- (A) A rubber ball and a stone of the same size.
- (B) A bicycle and a train.
- (C) A five rupees coin and a one-rupee coin.

Ans. Inertia depends on the mass of the object; an object with greater mass has greater inertia. The following objects have greater inertia:

- (A) Stone
- (B) Train
- (C) Five-rupee coin

- 11. Determine the time required by an object having weight 10 kg to gain the velocity of 80 m/s by exerting force of 50 N.**

Ans. Given

Force, $F = 50 \text{ N}$

Mass, $m = 10 \text{ kg}$

Velocity, $v = 80 \text{ m/s}$

Force, $F = ma$

As we know,

$$a = \frac{F}{m}$$

$$a = \frac{50}{10} = 5 \text{ m/s}^2$$

Now by the first equation of motion,

$$v = u + at$$

$$80 = 0 + 5t$$

$$t = \frac{80}{5} = 16 \text{ s}$$

LONG ANSWER Type Questions (LA)

[4 & 5 marks]

- 12. Two trucks of mass equal to 5000 kg are moving on a road. One is moving with a velocity of 50 m/s while the other is moving with a velocity of 2 m/s. Calculate the force required to stop them in 20 seconds. Which one of them requires more force and why?**

[Diksha]

Ans.Truck 1:

Mass, $m = 5000 \text{ kg}$

Initial velocity, $u = 50 \text{ m/s}$

Final velocity, $v = 0 \text{ m/s}$ (as it comes to rest)

Time taken by the truck to come to rest, $t = 20 \text{ s}$

To find the acceleration of truck 1 use the first equation of motion $v = u + at$

Now, substituting the given values in the above equation, we get,

$$0 = 50 + a \times 20$$

$$a = \frac{-50}{20}$$



$$a = -2.5 \text{ m/s}^2$$

A negative sign shows retardation.

As we know,

$$F = ma$$

$$F = 5000 \times 2.5$$

$$F = 12,500 \text{ N}$$

Truck 2:

As per the given data,

Mass = 5000 kg

Initial velocity = 2 m/s

Final velocity = 0 m/s (as it comes to rest)

Time taken by the truck to come to rest = 20 s

To find the acceleration of the truck, we use the first equation of motion

$$v = u + at$$

Now substituting the given values in the above equation, we get,

$$0 = 2 + a \times 20$$

$$20a = -2$$

$$a = \frac{-2}{20}$$

$$a = -0.1 \text{ m/s}^2$$

Negative sign shows retardation.

As we know, $F = ma$

$$F = 5000 \times 0.1$$

$$F = 500 \text{ N}$$

Hence, the force required to stop truck 1 is 12,500 N.

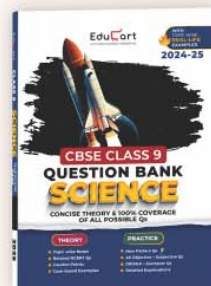
The force required to stop truck 2 is 500 N.

Therefore, we can conclude that the force required to stop truck 1 is greater than truck 2.



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